



About Team

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» HACK «
SYNAPSE

LINK TO PITCH VIDEO:

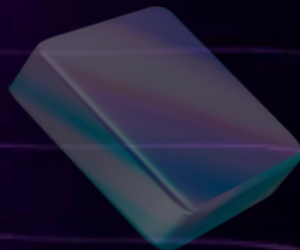
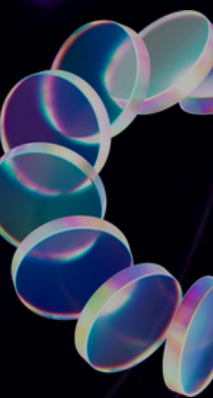
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Problem Description

THE \$21 TRILLION INFRASTRUCTURE CAPITAL CRISIS

- **Severe Payment Delays:** The average construction payment takes **80 to 90 days** to process, crippling small-business working capital.
- **Trapped Application Deposits:** Contractors must lock up thousands of dollars in "bid bonds" just to apply for public projects. Cities take **30 to 90 days** to return deposits to losing bidders.
- **Middleman Cash-Float Abuse:** Main contractors hold project funds in private bank accounts for months to earn interest or cover debts from unrelated projects.
- **Unprotected Retainage Savings:** Main contractors hold back **10% of every invoice** until the entire project finishes years later. If the main contractor goes bankrupt, workers lose their earned savings completely.
- **Why Legacy Software Fails:** Alternative tools track invoices and PDFs, but do not control the flow of money. The cash stays inside the **main contractor's bank account**.

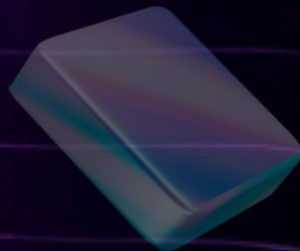




Solution

InfraFlow – Neutral Digital Lockbox for Construction Payments

- **Non-Custodial Smart Vault:** Replaces private corporate bank accounts with a neutral, code-governed treasury protocol.
- **Instant Application Refunds:** Automatically returns 100% of bid deposits to unselected bidders the second a contract is awarded.
- **Direct-to-Worker Payouts:** Milestone draws route 90% of funds directly to workers and subcontractors, bypassing middleman bank accounts.
- **Bankruptcy-Proof Retainage:** Automatically diverts 10% into a timelocked vault that is legally and cryptographically isolated from main contractor creditors.
- **Just-In-Time (JIT) Rolling Escrow:** Owners pre-fund only the active milestone (e.g., \$50k for Foundation), allowing traditional bank credit lines to be drawn phase-by-phase.





Flow of solution

1. RFP & Deposit Phase:

- * Bidders deposit application bonds into the InfraFlow smart contract.
- * Project owner selects winning contractor >> losing bidders receive instant refunds (< 5 seconds).

2. Milestone & Inspection Phase:

- * Contractor completes phase (e.g., Foundation / Framing).
- * Certified Inspector logs an IPFS report and approves via a Soulbound License Credential.

3. Automated Settlement Phase:

- * Smart contract auto-splits approved milestone funds:
- * 90% transfers directly to contractor/subcontractor wallets.
- * 10% locks into the 90-day Retainage Vault.

4. Final Release Phase:

- * 90-day post-completion timer expires with no defect claims >> 100% of retained savings automatically disburse.

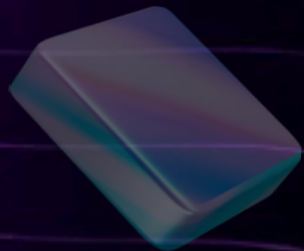




Technology Used

Lightweight, High-Velocity Web3 Architecture (TENTATIVE**)

- **Smart Contracts:** Solidity (built & tested via Foundry) for gas-optimized execution.
- **Token Standard:** ERC-20 (Mock USDC) for stable value transfers + OpenZeppelin access control modules.
- **Target Network:** Arbitrum / Base Sepolia (EVM Layer-2)** for sub-second confirmation times and near-zero gas costs.
- **Frontend Application:** Next.js (React) + Tailwind CSS for a clean, role-based dashboard.
- **Web3 Integration:** wagmi + viem + RainbowKit for seamless browser wallet connectivity.
- **Data Verification:** IPFS (Pinata) to store tamper-proof inspection reports and site photos off-chain.





Impact & Feasibility

Unique Selling Proposition (USP)

- **True Neutral Settlement:** Unlike Oracle/Procore, InfraFlow holds and settles funds programmatically without middleman custody.
- **Bankruptcy Isolation:** Retainage funds stay safe inside the smart contract, even if the main contractor collapses.

Measurable Impact

- **Collapses Payment Cycles:** Reduces payment wait times from 90 days to under 5 seconds.
- **Unfreezes Capital:** Instantly returns millions in trapped bidding deposits back to local businesses.
- **Eliminates Legal Frictions:** Stops subcontractor mechanics' liens caused by non-payment.

Operational Feasibility

- **Requires zero hardware changes:** inspectors and owners approve draws using standard mobile web browsers.
- **Phase-by-phase funding:** aligns with standard commercial bank construction loan draws.



Prototype (PRESENTABLE PROTOTYPE IS NOT AVAILABLE YET !!!)





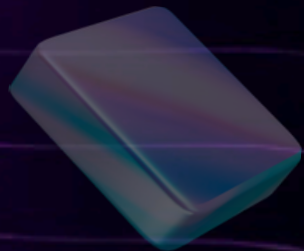
Future scope

Multi-Sig Inspector Oracles:

**Integrating 3-of-3 sign-offs (Architect + Site Inspector + City Official).
ERP API Connectors Bi-directional sync with traditional software like
Procore and QuickBooks for automated accounting.**

Fiat Off-Ramps:

**Integration with localized bank off-ramps (e.g., Stripe / Transak) for
direct-to-bank fiat conversion.**





Submission Guidelines

- Use the official PPT template only.
- Submit the presentation in PDF format.
- Upload a 1.5–2 minute video pitch via Google Drive.
- Set Drive access to "Anyone with the link can view."
- Attach the video pitch Google Drive link in the PPT.
- Verify all submitted links before submission.
- Only the Team Leader can submit the final entry.
- One submission per team. No edits after submission.

Evaluation Criteria

- Innovation & Originality
- Problem Understanding
- Technical Approach
- Feasibility & Scalability
- Presentation & Communication
- Potential Impact



THANK YOU!

WE LOOK FORWARD TO YOUR INNOVATIVE IDEAS.

BEST OF LUCK!

IEEE HACK SYNAPSE 2026

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